BlueFairy Anti-Spam

Installation & Features Guide

Invisible spam protection for WordPress & WooCommerce.
No CAPTCHAs. No friction. Zero configuration required.

ALWAYS FREE

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Requirements & Installation

Requirement	Minimum
PHP	8.1+
WordPress	6.5+
WooCommerce	7.0+ (optional)

1. Upload the `bluefairy-antispam/` folder to `wp-content/plugins/`
2. Activate via **Plugins > Installed Plugins**
3. Done — honeypot and time trap are active immediately

Optional configuration: **Tools > Anti-Spam**

Honeypot Trap

A hidden input field is added to every supported form. The field is positioned 9,999 pixels off-screen and marked `aria-hidden` — completely invisible to real users. Bots that blindly fill all form fields trigger the trap and are blocked before their submission reaches WordPress.

The field name is drawn from a pool of eight realistic-sounding names (`website2` , `address_confirm` , `phone_verify` , etc.), chosen deterministically per form type, so bots cannot blacklist a single known field name.

Forms covered

- WordPress comments (traditional POST + REST API)
- WordPress user registration
- WooCommerce product reviews
- WooCommerce My Account registration
- WooCommerce shortcode checkout

Time Trap

At page load, a cryptographically signed HMAC-SHA256 token is embedded in a hidden field. The token contains a nonce, a timestamp, and the form type. When the form is submitted, the plugin verifies the signature and checks how much time has elapsed.

Submissions that arrive in under 3 seconds (configurable in settings) are rejected — bots submit instantly; humans take at least a few seconds to read and fill a form. Tokens expire after 24 hours.

Security properties

- HMAC-SHA256 — the token cannot be forged without the site's secret key
- Secret key generated at activation via `random_bytes(32)` , stored non-autoloaded
- Form type embedded in the token — a token for `wp_comment` cannot be replayed on `wc_checkout`

Stop Forum Spam (optional)

Stop Forum Spam maintains the world's largest database of known spammer IPs and email addresses, crowdsourced from thousands of websites.

This feature is disabled by default. Enable it under **Tools > Anti-Spam** only after updating your privacy policy to disclose that visitor IPs and emails are sent to a third-party service.

How it works

- On each protected form submission, the visitor's IP and email are sent to `api.stopforumspam.org/api`
- Results are cached for 7 days (hit) or 1 hour (clean) to avoid redundant API calls
- A circuit breaker skips SFS after 5 consecutive API failures and resets after 10 minutes
- You can report blocked visitors back to SFS with a free API key

Confidence threshold

Default: 1. Set higher (e.g. 50) to only block high-confidence matches. Set to 0 to block any match in the database.

Admin Log

Every blocked submission is logged to `wp_bfas_log`. The log page is at **Tools > Anti-Spam Log**.

Column	Value
Date	UTC timestamp of the block
Trap	`honeypot`, `timetrap`, or `sfs`
Form	`wp_comment`, `wp_register`, `wc_review`, `wc_register`, `wc_checkout`, `wc_checkout_block`
IP Address	Visitor's IP
Email hash	HMAC-SHA256 of the email — raw address never stored
Reported to SFS	Whether this block was reported back to Stop Forum Spam

Log entries are automatically deleted after 90 days (configurable). Deletion runs in daily batches of 1,000 rows to avoid timeout.

The log page supports: search by IP, filter by trap type, sort by date or trap, bulk delete.

Dashboard Widget

The **Anti-Spam Activity** widget on the WordPress Dashboard shows:

- Total spammers blocked (all time)
- Blocked this week / this month
- Breakdown by trap type

Settings Reference

All settings under **Tools > Anti-Spam**.

Setting	Default	Description
Honeypot	On	Toggle the honeypot trap globally
Time trap	On	Toggle the time trap globally
Min submit time	3 seconds	Submissions arriving faster are blocked (1–120 s)
Enable SFS lookup	Off	Opt-in to Stop Forum Spam checking
SFS API key	Empty	Required only for reporting; leave blank to check only
SFS threshold	1	Block if SFS confidence ≥ this value (0–100)
Log retention	90 days	How long to keep log entries

WooCommerce Notes

Shortcode checkout

Full protection — honeypot and time trap fields are injected via the `woocommerce_checkout_after_order_notes` hook. The submission is validated via `woocommerce_checkout_process`.

Block checkout

The WooCommerce Block checkout submits via the Store API (JSON), not an HTML POST form, so honeypot and time trap fields cannot be injected. Only the SFS check applies (if enabled). This is a platform limitation, not a bug.

Logged-in users

All traps are bypassed for logged-in users — no friction for real customers.

Frequently Asked Questions

Will this break my checkout?

No. Real customers never interact with any trap field. The honeypot is invisible. The time trap gives 24 hours before expiry — far longer than any checkout takes.

Do I need a Stop Forum Spam account?

No. SFS is optional and off by default. Honeypot and time trap work without it.

What gets stored?

When a submission is blocked: the time, which trap fired, which form, the IP address, and a hashed version of the email. The raw email is never stored.

Does it conflict with other security plugins?

No. It hooks into standard WordPress and WooCommerce filter/action points and doesn't modify any global state.

License & Support

GPL-2.0-or-later. Always free.

Support: bluefairydevelopment.com

Bug reports and contributions: github.com/bluefairydevelopment/bluefairy-antispam